

MARYUM IRSHAD

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PhD Candidate, Computational Biochemistry | Molecular Dynamics | Protein Biophysics | Machine Learning

PhD candidate in Computational Biochemistry specializing in computational biophysics, atomistic molecular dynamics simulations, protein structure and dynamics, and machine-learning approaches for biomolecular interactions. Experienced in GROMACS-based simulations, protein-protein binding energetics, structural bioinformatics, Python/PyTorch development, and automated computational workflows.

STRENGTHS AND EXPERTISE

Computational Biophysics: Molecular dynamics simulations, protein structural dynamics, protein-protein and protein-ligand interactions, binding-energy analysis

Molecular Modeling & Simulation: GROMACS, VMD, PyMOL, UCSF ChimeraX

Structural Bioinformatics: Protein structure analysis, conformational analysis, and interpretation of biomolecular simulation data

Machine Learning & AI: PyTorch, graph-based machine learning, NetworkX, PCA, machine-learning model development

Scientific Programming: Python, Biopython, RDKit, data analysis, scripting, and workflow automation

Computational Workflows: Pipeline Pilot, automated data-processing pipelines, large-scale molecular and structural data analysis

EDUCATION

Ph. D. Chemistry

University of Cincinnati

CGPA: 3.95

2022 - Present

Bachelors of Science in Chemical Engineering, Honors

University of Engineering and Technology, Lahore

CGPA: 3.89

2011 - 2015

RESEARCH EXPERIENCE

University of Cincinnati

Graduate Research Student

August 2022 - Present

Conduct computational research investigating protein structure, dynamics, energetics, and regulatory mechanisms using molecular dynamics simulations, structural bioinformatics, and machine-learning approaches.

Key Highlights:

- **Developed FlexENN, an Interface-focused Message-Passing Graph Neural Network**, to predict protein-protein binding energies from structural representations of biomolecular interfaces, especially for intrinsically disordered protein complexes.
- **Investigated the Allosteric Regulation of AAA+ microtubule severing enzyme, Katanin**, integrating molecular simulations, bioinformatics, structural analysis, and machine learning techniques, advancing the understanding of its function and regulatory mechanisms.
- **Studied MAP7-Microtubule Interactions Across Different Microtubule Isoforms**, using Dynamic Cross-Correlation Matrices, RMSF, Principal Component Analysis (PCA), and structural comparisons to characterize protein dynamics.
- Analyzed protein-protein and protein-ligand interfaces, conformational changes, residue-level interactions, and structural features across simulated biomolecular systems.
- **Developed Python Workflows** for automated processing, analysis, and visualization of molecular dynamics trajectories and protein structural data.
- Executed computationally intensive molecular simulations and analyses using **Linux GPU environments**.

Procter & Gamble at UC Digital Accelerator

May 2025 - Present

Graduate Research Assistant – Human Safety

- Develop **Python-based computational protocols for structure-based analogue identification** to support chemical safety assessment using **Read Across methodologies**.
- Develop automated approaches for identifying chemically relevant analogues from molecular datasets using structural similarity and molecular transformation criteria.
- Apply cheminformatics and data-processing tools, including **Python, RDKit, Pandas, and Pipeline Pilot**, to automate chemical structure analysis and analogue-selection workflows.
- Investigate **matched molecular pair approaches** to identify structurally meaningful relationships between compounds for human-safety applications.

University of Engineering and Technology, Lahore

October 2019 - July 2022

Research Assistant, HEC-NRPU Project

- Synthesized supported metal catalysts using impregnation and co-precipitation techniques. Characterized catalytic materials using **XRD, BET, and FTIR**. Evaluated catalytic performance for glycerol conversion and analyzed products using **gas chromatography (Shimadzu GC-2014)**. Commissioned and operated laboratory equipment including a Parr bench-top reactor, Carbolite tube furnace, and Karl Fischer titrator. Developed laboratory operating and safety procedures and supported laboratory maintenance and experimental workflow management.

PUBLICATIONS

Irshad, M., Walpalage, K., Zhaocheng, Z., Gillen M., George, S., Dima, R.I.: *Probing Mechanisms of Allosteric Regulations in AAA+ ATPases for Microtubule Severing and Protein Disaggregation*. (Submitted to JCP)

Irshad, M., Ori M. K., Dima, R.I.: *Leveraging Machine Learning Models to Predict Protein-Protein Binding Energies for Globular and Intrinsically Disordered Proteins*. (Submitted to JCIIM)

Stump, E., **Irshad, M.**, Dima, R.I.: *The differential effects of MAP7 and Tau on Microtubules Dynamics and Stability*. (Manuscript in progress)

Shakur, M., **Irshad, M.**, Haider, M. S.: *Hydrogenolysis of Glycerol to Propanediols over Ni Supported on MgO Catalyst. Pertanika Journal of Science & Technology*, **33**(2), 2025.

SELECTED PRESENTATIONS

Poster Presentation

“Automating Structure Based Analogue Selection for Read Across: The Power of Matched Molecular Pairs”, *SOT Meeting*, 2026

“A Machine Learning Approach to Extracting Protein-Protein Binding Energies From Molecular Dynamic Simulations”, *Biophys. Soc. Meeting*, 2026

“From Small Tests to Big Data: Calculating Protein-Protein Binding Energy Using Machine Learning”, *Oesper Symposium*, October 2025

“A Computational Approach to Understand MAPs”, *Oesper Symposium*, October 2024

“Computational Methods to Predict Allostery in Katanin Monomer”, *Oesper Symposium*, October 2023

AWARDS & HONORS

Student Research Achievement Award for outstanding presentation at Biophysical Society Annual Meeting **2026**

Procter & Gamble Poster Competition for clearly communicating, innovative research presented during the Oesper Symposium **2025**

Hillstorm Travel Award as an outstanding graduate student in support of a presentation before a scientific audience

IRiS Graduate Research Fellowship **August 2024 – April 2025**

University of Cincinnati – Great American Insurance Group

Conducting an interdisciplinary research project aimed at understanding the perception of telematics among small commercial fleet companies and insurance agents. Analyzing industry trends, developing insights, and contributing to understanding the implications of telematics on insurance policies and practices

Chemistry Doctoral Enhancement Fellowship to an outstanding graduate student to support recent accomplishments **2024**

Henry Hochstetter Prize for excellent performance as a graduate teaching assistant **2023**

TEACHING AND INDUSTRIAL EXPERIENCES

Graduate Teaching Assistant **August 2022 – May 2025**

Taught **Organic Chemistry I, General Chemistry I and II** (lab and recitations)

Treasurer, Chemistry Graduate Student Association **2023 - 2024**

Administering and documenting the expenses for the student association, along with planning and executing events

Research Experience for Undergraduates Co-Coordinator **Summer 2023**

Collaborated in planning and executing events for the students as well as end of summer presentations

Fauji Fertilizers Company, Mirpur Mathelo, Pakistan **October 2016 – July 2018**

Junior Executive Production (Ammonia)

Led a team of 8 staff members in the Ammonia Unit, ensuring optimal plant performance and safe operation. Managed plant emergencies, supervised plant turnaround activities, and coordinated the safe shutdown/start-up and maintenance procedures, including handling Management of Change (MOC) processes.

Fauji Fertilizers Company, Mirpur Mathelo, Pakistan **September 2015 – September 2016**

Management Trainee Production (Ammonia)

Gained comprehensive knowledge of Ammonia plant operations through hands-on training, including emergency handling procedures, technical aspects of plant systems, and equipment maintenance. Completed specialized training in problem-solving, decision-making, and various technical systems like SAP-ERP and Cathodic Protection.

Engro Fertilizers Company, Daharki, Pakistan **July 2014 – August 2014**

Intern, New Venture Division

Conducted research on alternative feed and fuel sources for ammonia production and explored the potential of shale gas versus natural gas. Gained an overview of ammonia and urea plant operations and studied exploration and production companies in Pakistan.

PROFESSIONAL SERVICES & LEADERSHIP

- **Reviewer**, IEEE Biosensors, 2026
- **Judge**, Ohio State Science Day 2026
- **Reviewer**, Student presentations at the 8th Annual University of Cincinnati Undergraduate Scholarly Showcase and Research Symposium
- **Judge**, 4th Annual Greater Cincinnati Regional High School Ethics Bowl
- **Event Manager**, Graduate Student Government of University of Cincinnati, 2023 – 2026
- **Treasurer**, Chemistry Graduate Student Association, 2023–2024
- **Research Experience for Undergraduates Co-ordinator**, Summer 2023